# A Vendor-Ready Data Quality Framework for Reliable Healthcare Analytics

Kyle Fuller

M.S. Healthcare Informatics | University of Central Florida | School of Global Health Management and Informatics

Faculty Mentors: Amanda Walden, PhD, RHIA, CHDA; Michelle Crozier, PhD, MPH, CPH

UCF STUDENT RESEARCH WEEK

## Background & Objective

Healthcare organizations rely on dashboards and reports to support operational, financial, and clinical decision-making. However, reporting is only as reliable as the underlying data. Common data quality issues such as duplicate records, missing values, invalid dates, and broken relationships between tables can distort counts, rates, and trends.

**Research Question: Can a simple SQL-based validation framework detect common healthcare data quality issues before they affect reporting outputs?**

**Objective: To design and test a vendor-ready SQL-based data quality framework using a simulated healthcare reporting dataset and evaluate whether common data quality issues can be identified before reports are produced.**

## Methods & Simulated Dataset

A simulated relational dataset was built in DB Browser for SQLite using three linked tables. SQL checks were run to detect duplicate IDs, missing values, invalid dates, broken table relationships, and missing diagnosis codes. No real patient data were used.

**patients**

patient_id
patient_name
dob
gender

**encounters**

encounter_id
patient_id
admit_date
discharge_date
provider_name

**diagnoses**

diagnosis_id
encounter_id
diagnosis_code

## Validation Rules Used

- Duplicate patient IDs
- Duplicate encounter IDs
- Missing patient IDs in encounters
- Invalid or future encounter dates
- Diagnoses records without matching encounters
- Missing diagnosis codes

**Representative SQL**

```
SELECT encounter_id, COUNT(*) AS duplicate_count
FROM encounters
GROUP BY encounter_id
HAVING COUNT(*) > 1;
```

## Framework

Raw Healthcare Data → SQL Validation Rules → Error Detection → Flag / Correct Invalid Records → Clean Reporting Dataset → Reliable Analytics and Reporting

Validate first, report second. The workflow places data quality checks before dashboard or report delivery so reliability problems are detected upstream.

## Results

| Patients reviewed | Encounters reviewed | Encounter issue rate |
|---|---|---|
| 4 | 5 | 80% |

**Figure 1. Validation errors detected**

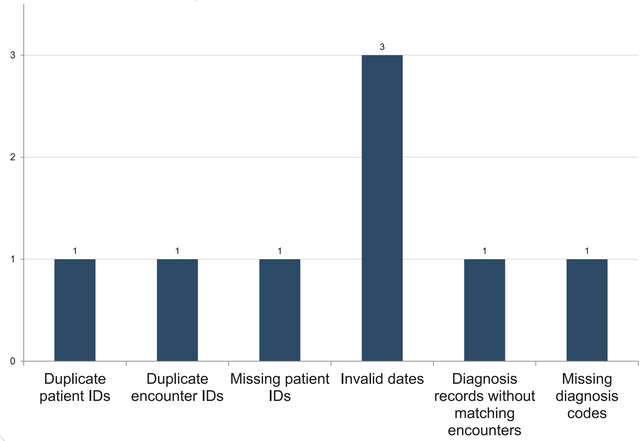


**Table 1. Results summary**

| Validation Rule | Errors Found |
|---|---|
| Duplicate patient IDs | 1 |
| Duplicate encounter IDs | 1 |
| Missing patient IDs | 1 |
| Invalid dates | 3 |
| Diagnosis records without matching encounters | 1 |
| Missing diagnosis codes | 1 |

The SQL-based validation framework identified six distinct types of data quality issues in the simulated healthcare dataset. One duplicate patient ID, one duplicate encounter ID, one missing patient ID, three invalid encounter dates, one diagnosis record without a matching encounter, and one missing diagnosis code were detected. Overall, 4 of 5 encounter rows failed at least one critical validation check, demonstrating how data quality problems can materially reduce trust in downstream healthcare reporting.

**Key takeaway: Even across a small simulated dataset 4 of 5 encounter rows (80%) failed at least one key validation check, showing why data quality issues can distort later dashboards and reports.**

## Conclusion & Implications

1. Basic SQL validation checks can identify high-risk data quality issues before reports or dashboards are built.
2. Even in a small simulated dataset, duplicate, missing, invalid, and broken records were common enough to reduce trust in reporting outputs.
3. For healthcare analysts and vendors, the framework functions as a repeatable pre-reporting control rather than a one-time cleanup exercise.
4. This project does not introduce new data quality checks; it adapts established validation concepts into a lightweight, repeatable pre-reporting workflow for operational analytics.

## Future Work

1. Expand to claims, enrollment, provider directory, or revenue cycle datasets
2. Add provider-name standardization checks
3. Automate validation inside ETL and reporting workflows
4. Feed validation outputs into ongoing monitoring dashboards
5. Define severity thresholds and appropriate criteria before dashboard delivery

## Selected References

1. Office of the National Coordinator for Health Information Technology. HealthIT.gov.
2. Centers for Medicare & Medicaid Services. CMS Provider Data and Hospital Data resources.
3. Agency for Healthcare Research and Quality. AHRQ Quality Indicators.
4. Centers for Disease Control and Prevention. National Center for Health Statistics resources.
5. American Medical Informatics Association. AMIA resources.

## Acknowledgements

1. Amanda Walden, PhD, RHIA, CHDA
2. Michelle Crozier, PhD, MPH, CPH
3. Hannah Nguyen, MHA
4. UCF School of Global Health Management and Informatics
5. UCF Student Research Week 2026

*Simulated dataset created for demonstration only; no real patient data were used.*